

program2.py X

program2.py

```
1 processes = [  
2     {"pid": "P1", "arrival": 0, "burst": 7},  
3     {"pid": "P2", "arrival": 2, "burst": 4},  
4     {"pid": "P3", "arrival": 4, "burst": 1},  
5     {"pid": "P4", "arrival": 5, "burst": 4},  
6 ]  
7  
8  
9 def fcfs(process_list):  
10     current_time = 0  
11     intervals = []  
12  
13     ordered = sorted(  
14         process_list,  
15         key=lambda p: (p["arrival"], p["pid"])  
16     )  
17  
18     for process in ordered:  
19         if current_time < process["arrival"]:  
20             intervals.append(("IDLE", current_time, process["arrival"]))  
21             current_time = process["arrival"]  
22  
23         start = current_time  
24         end = start + process["burst"]  
25  
26         intervals.append((process["pid"], start, end))  
27         current_time = end  
28  
29     return intervals  
30  
31  
32 def sjf(process_list):  
33     remaining = process_list.copy()  
34     current_time = 0  
35     intervals = []  
36  
37     while remaining:  
38         ready = [  
39             p for p in remaining  
40             if p["arrival"] <= current_time  
41         ]  
42  
43         if not ready:  
44             next_arrival = min(p["arrival"] for p in remaining)  
45             intervals.append(("IDLE", current_time, next_arrival))  
46             current_time = next_arrival
```

```
continue
```

```
process = min(  
    ready,  
    key=lambda p: (p["burst"], p["arrival"], p["pid"])  
)
```

```
start = current_time  
end = start + process["burst"]
```

```
intervals.append((process["pid"], start, end))  
current_time = end  
remaining.remove(process)
```

```
return intervals
```

```
def show_result(title, intervals):
```

```
    print("\n" + title)  
    print("Process  Start  End")
```

```
    sequence = []
```

```
    for pid, start, end in intervals:  
        print(f"{pid:<9} {start:<7} {end}")
```

```
        if pid != "IDLE":  
            sequence.append(pid)
```

```
    print("Sequence:", " " -> ".join(sequence))
```

```
print("INPUT PROCESSES")
```

```
print("PID  AT  BT")
```

```
for p in processes:
```

```
    print(f'{p["pid"]:<5} {p["arrival"]:<4} {p["burst"]}')  
)
```

```
show_result("FCFS SCHEDULING", fcfs(processes))
```

```
show_result("SJF SCHEDULING", sjf(processes))
```

```
PS Microsoft.PowerShell.Core\FileSystem::\\wsl.localhost\Ubuntu-24.04\home\shashi_linux\OS-Second-program> & 'C:\Python313\python.exe' 'c:\Users\user1.SHASHIRAY.000\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '55917' '--' '\\wsl.localhost\Ubuntu-24.04\home\shashi_linux\OS-Second-program\program2.py'
```

INPUT PROCESSES

PID	AT	BT
P1	0	7
P2	2	4
P3	4	1
P4	5	4

FCFS SCHEDULING

Process	Start	End
P1	0	7
P2	7	11
P3	11	12
P4	12	16

Sequence: P1 -> P2 -> P3 -> P4

SJF SCHEDULING

Process	Start	End
P1	0	7
P3	7	8
P2	8	12
P4	12	16

Sequence: P1 -> P3 -> P2 -> P4

```
PS Microsoft.PowerShell.Core\FileSystem::\\wsl.localhost\Ubuntu-24.04\home\shashi_linux\OS-Second-program>
```